

Microservices: A definition of this new architectural term

Source: J. Lewis and M. Fowler "Microservices: A definition of this new architectural term," 2014. [Online]. Available: <https://martinfowler.com/articles/microservices.html>

Rating: ★★★★★

Own Keywords: Background, Domain Driven Design (DDD), Microservices, Migration, Monolith

Reading Progress: Third Pass

Introduction (What is the Paper about?)

- The authors wish to describe a new up-coming trend in 2014 towards Microservices
- The authors seek to describe what they have seen is typical for microservices
 - they compare microservices to the old fashioned monolith and point out different key features and ways of thinking

Methods

- the authors describe their observations and experiences in industry applying no further or special methodology

Scientific Benefit (*Here the observations*)

- **The authors provide several useful definitions:**
 - **Component:** *"A Component is a unit of software that is independently replaceable and upgradeable."*
 - **Libraries:** *"We define libraries as components that are linked into a program and called using in-memory function calls."*
 - **Service:** *"Services are out-of-process components who communicate with a mechanism such as a web service request, or remote procedure call."*
 - are independently deployable - libraries not
 - **Microservices:** *"In short, the microservice architectural style 1 is an approach to developing a single application as a suite of small services, each running in its own process and communicating with lightweight mechanisms, often an HTTP resource API. These services are built around business capabilities and independently deployable by fully automated deployment machinery. There is a bare minimum of centralized management of these services, which may be written in different programming languages and use different data storage technologies."*
 - seeks to build applications as suites of services
 - services seek to be independently deployable and therefore scalable
 - each service may use different programming languages and technology stacks to fit their central business capability the best
- **Enterprise Applications often consists of three main parts:**
 1. User Interface (UI)
 2. Server-side application
 3. Database (DB)
- **Monoliths are built as one single deployable unit**
 - all logic for handling a request runs just in a single process
 - any (even small) changes involve building and deploying a new version of the server-side application

- people are facing frustrations with monoliths as applications are more deployed to the cloud
- **Monoliths are horizontally scalable** by deploying multiple instances of the monolith itself rather
 - scaling of single parts of the monolith is not possible
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- **Monoliths struggle over time to maintain a good modular structure**
 - using just the provided features of the used programming language to divide and structure the application
 - hard to keep changes being only limited to one module/place in code
 - often affects many places in the code
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Challenges / Future Work

- authors are active members of the microservice community
 - potentially more subjective in this regard

Limitations

Paper Outline and Read Progress

1. [X] Introduction: Definition of microservices and how it is different from monoliths
2. [X] Characteristics of a Microservice Architecture

Own Comments

- (+) provides several basic ideas and definitions
- (+) key article of microservices
- (-) old
- (-) statements are based on observations